Mines Paris PSL - Etilab

Avoiding the British

Brexit and Partner Choice in European Venture Capital

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Outline

A shock that never regulated syndication

- Syndication is relational: a co-investment tie is an asset whose value is the deal flow it generates (Hochberg et al., 2007, 2010).
- Before June 2016, UK VCs were among the most sought-after partners in Europe.
- **23 June 2016.** 52% vote to leave, largely unanticipated (Breinlich et al., 2018); sterling -8% overnight; UK EPU at an all-time high.

The point of departure

Nothing in the withdrawal restricted co-investment. A French fund could write a cheque alongside a British one on 22 June 2016 and today — identical terms, identical formalities.

This paper documents that they stopped doing so anyway.

▸ The AIFMD passport

This paper

Research question

Did the referendum change *whom* continental venture capitalists pick as partners — and on which margins did it stop?

Findings

- **One effect.** -8.2 pp on a base of 18.7% ($p < 0.001$): a relative decline of 44%, dated on the referendum, still present in 2022–25.
- **Four boundaries.** Ties hold ($+0.3$ pp, $p = 0.77$); the corridor holds ($[-20\%, +26\%]$); activity, stages, geography, amounts, follow-ons, exits null; the British side not identified.
- **No mechanism asserted.** Four candidates remain consistent.

The five results

	Outcome	Estimate	95% CI	Verdict
Partner choice	UK-partner rate (pp)	-8.20	$[-13.11, -3.15]$	**Identified**
Dyads	Tie survival, triple diff. (pp)	$+0.30$	$[-1.84, +2.51]$	Precise null
Corridor	Joint deal volume (log pts)	$+0.03$	$[-0.20, +0.26]$	Precise null
Activity	$P(\text{active})$, window $[0, 6]$ (pp)	$+3.18$	$[-1.47, +8.16]$	Not identifi
UK side	Mirror estimate, British panel (pp)	-6.01	$[-11.52, -0.33]$	Not identifi

Wild-cluster bootstrap-t intervals throughout. The two precise nulls exclude economically meaningful effects. “Not identified” is *not* “zero” — the last row’s interval excludes zero — and the distinction is the subject of Section 5.

Precision

Literature

- **Syndication as relational capital.** (Lerner, 1994; Brander et al., 2002; Hochberg et al., 2007, 2010; Sorenson and Stuart, 2001)
  - A tie is a durable, rivalrous asset; partner choice sets the future opportunity set.
- **Certification.** (Megginson and Weiss, 1991; Hsu, 2004; Nahata, 2008; Hellmann and Puri, 2002)
  - If the British were certifying, their exit should degrade outcomes. It does not.
- **Cross-border VC and its frictions.** (Cumming and Dai, 2010; Tykvová and Schertler, 2012; Guiso et al., 2009; Bottazzi et al., 2016)
  - Bilateral trust supports cross-border investment — the strongest prior for our result.
- **Brexit and finance.** (Breinlich et al., 2018; Born et al., 2019; Bosio et al., 2025)
  - Bosio et al.: flows and geography at the hub level. Ours: partner choice, firm level.
- **Inference with few clusters.** (Cameron et al., 2008; MacKinnon and Webb, 2017; MacKinnon et al., 2023)

Data

Source. Crunchbase — investment-level panel: each round $\times$ each participating investor, with HQ countries.

Scope. Venture rounds, OECD targets. $1{,}211{,}489$ investor–round observations.

Time. The semester; event time τ relative to 2016H2. $\tau = -1$ is the referendum semester (never a reference), $\tau = -2$ (2015H2) the omitted category.

Continental Europe. European countries excluding the UK *and Ireland*.

- Ireland is the only country removed by name: the Common Travel Area survived, and Dublin took entities leaving London.
- Tested rather than decreed: adding it gives $448 \to 470$ investors, $-8.20 \to -8.01$ pp ($p < 0.001$).

▸ What Crunchbase does not see

Estimation sample

Continental VCs with ≥ 2 pre-2014 deals and ≥ 1 pre-2014 British co-investor

	High exposure	Low exposure
Investors	216	232
Pre-2014 deals (mean)	10.51	32.48
Pre-2014 deals with a UK partner (mean)	3.59	2.91
UK exposure share (mean)	0.429	0.108
Small (≤ 5 pre-2014 deals), %	58.8	5.2

The mechanical size gap. A given number of British partnerships is a larger share of a smaller book, so highly exposed investors are mechanically smaller (58.8% vs. 5.2% small).

$\Rightarrow$ Investor FE absorb the level, but any outcome correlated with size and trending over the sample would contaminate the estimate. Every specification therefore carries a small$_i \times$ post$_s$ control.

No untreated unit

The core difficulty

Brexit reaches every European venture capitalist on the same date. There is **no group the referendum did not touch**.

- A conventional difference-in-differences is unavailable.
- Pre/post cannot separate the referendum from the 2020–21 venture boom.

Identification by differential exposure. Among agents all treated, those whose pre-determined reliance on the affected channel was higher should respond more. Semester FE absorb the common component.

The estimand, stated explicitly

A *relative* effect. The level effect of Brexit on European VC is not recoverable — by this or any design on these data.

Specification

$$y_{is} \;=\; \sum_{\tau \neq -2} \beta_\tau \cdot \mathbf{1}\{s = \tau\} \cdot \text{High}_i \;+\; \gamma\,(\text{small}_i \times \text{post}_s) \;+\; \alpha_i \;+\; \delta_s \;+\; \varepsilon_{is}$$

- Investor and semester FE; High_i pre-determined, above the sample median.
- Clustering on the investor's *country*: G between 21 and 30.
- Wild-cluster bootstrap-t (Rademacher, equal-tailed, $B = 999$); CIs by inversion.
- Pre-trends by **restricted** bootstrap — the null imposed in the DGP — not by the asymptotic Wald statistic.

Why this matters

At these cluster counts the Wald test rejects parallel pre-trends where *no* pre-period coefficient has $|t| > 1.2$. With few clusters, it is the asymptotic version that misleads.

Early classification

The problem. Classifying on the full pre-referendum window invites regression to the mean: an unusually high UK share in 2015 mechanically reverts.

The fix. Exposure classified on deals before 1 January 2014 only; measurement panel starts in 2014.

$\Rightarrow \tau \in \{-5, -4, -3\}$ is a **held-out window**, used only to test parallelism.

Not a cosmetic correction

Full-window classification yields a -7.1 pp “divorce”, significant at all conventional levels — and -5.1 for France, -8.0 for Germany. Early classification removes all three.

Never test pre-trends on periods contained in the window that defines treatment: the flat gap is mechanical.

The falsification protocol (fixed *ex ante*)

One protocol, applied identically to the UK and to twelve placebo corridors:

1. the static exposure $\times$ post coefficient;
2. a joint pre-trend test on $\tau \in \{-5, -4, -3\}$;
3. the same test including $\tau = -1$, the referendum semester;
4. three post windows: $[0, 6]$, $[7, 11]$ (2020–21), $[12, 17]$ (2022–25).

The reading grid, also fixed before looking

Significant static coefficient, clean pre-trends on **both** tests, consistent sign in **all three** windows.

- Confined to 2020–21 $\Rightarrow$ a boom phenomenon.
- Absent from the first window $\Rightarrow$ not a referendum effect.

The UK faces the same grid as the placebos, and we report the verdict it returns.

Whom they stopped picking

Universe. Continental VCs with ≥ 2 pre-2014 deals and ≥ 1 pre-2014 British co-investor. *Every* investor in the sample is UK-exposed; the contrast is above vs. below the median exposure share.

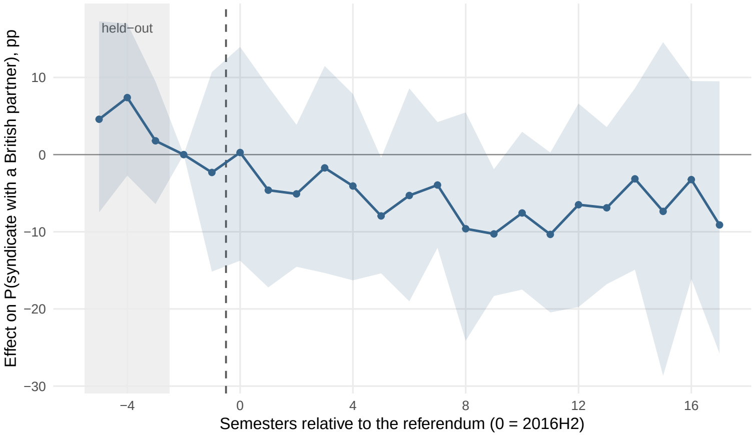
Outcome. huk$_{is}$ $= 1$ if investor i syndicated with a British investor in semester s.

Result

-8.2 percentage points ($p < 0.001$; 95% CI $[-13.1, -3.2]$) against a control base rate of 18.7% — a relative decline of **forty-four percent.**

- Present in all three windows: -6.4 in H2 2016–2019, -10.6 in 2020–21, -8.3 in 2022–25.
- Begins in the semester containing the referendum; does not accelerate when the law finally changes at end-2020.

Partner choice — event study



P(syndicate with a British partner), high vs. low pre-2014 exposure. Early classification, investor and semester FE, WCB-t bands ($B = 999$, country clusters). Ref. 2015H2; shaded band = held-out window. Pooled: -8.2 pp.

Two features we raise ourselves

1. The semester coefficients are imprecise. Only 2 of 18 post estimates have a bootstrap interval excluding zero.

$\Rightarrow$ Power, not identification: each semester is one draw from 448 investors. The effect is identified off the post-period average, which *is* precise.

2. The held-out coefficients are not flat. In 2014 the high group sits 4.6 and 7.4 pp *above* the low group, and the gap closes monotonically.

$\Rightarrow$ The joint tests pass ($p = 0.52$, $p = 0.70$), but such tests have low power (Roth, 2023). **We cannot exclude that part of the decline continues a drift already under way** — what argues against it is the placebo evidence, not the pre-trend test.

The thirteen-corridor battery

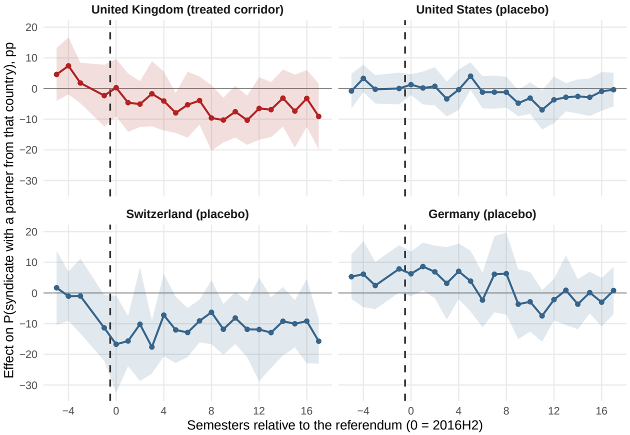
P (syndicate with a partner from the corridor country), static β in pp

Partner	Static	Pre-trends		Windows		Verdict
	(pp)	(i)	(ii)	$[0,6]$	$[12,17]$	
GBR	$\mathbf{-8.2^{***}}$	0.52	0.70	-6.4^{**}	-8.3^{**}	**Clears the grid**
CHE	-9.2^{***}	0.98	0.83	-10.9^{***}	-9.2^{***}	Clears the grid
USA	-2.1	0.18	0.31	-0.3	-2.7	null
CAN	$+0.5$	0.28	0.34	-2.1	$+4.2$	null
ISR	-0.7	0.52	0.09	$+0.8$	-1.3	null
JPN	-0.8	0.78	0.64	$+0.3$	$+1.8$	null
KOR	-4.6	0.37	0.06	-3.0	-4.0	null
FRA	-3.0	0.56	0.68	-1.3	-5.0	null
DEU	-3.0^{**}	0.64	0.47	$+0.4$	-5.5^{***}	mixed profile
NLD	-0.8	0.48	0.71	$+1.5$	-0.1	null
SWE	-3.2	0.01	0.07	-3.2	-2.8	pre-trends fail
BEL	-5.2^{**}	0.05	0.05	-1.3	-6.6^{*}	pre-trends fail
Non-EU OECD	-1.5	0.24	0.43	-0.7	-1.5	null

Pre-trend (i): joint test on $\tau \in \{-5,-4,-3\}$; (ii): additionally including $\tau = -1$. Both by restricted WCB. Verdicts assigned mechanically by the ex-ante grid. $^{***}p < .01$, $^{**}p < .05$, $^{*}p < .10$.

$\Rightarrow$ **Eleven corridors return nothing. Two clear the grid: the UK and Switzerland.**

Switzerland breaks *before* the vote



Four corridors, identical axes and specification. The Swiss corridor has already collapsed by $\tau = -1$ — the campaign semester, before the vote — then plateaus (-11.4, $t = -2.1$). The German and American corridors move only from 2020. Only the British corridor declines from the referendum onwards and stays down.

What the timing does and does not settle

- The static filter admits **two** corridors; the timing admits **one**.
- No channel running from a referendum on Union membership can produce a break that *precedes* the vote — which is the Swiss profile.

The concession we make explicitly

A reader unwilling to weigh timing should conclude that the design identifies a British *and* Swiss decline and cannot separate them — the claim intact in form, weaker in specificity.

We report this rather than tighten the grid until one corridor survives.

Where the effect stops

The shift in partner choice is *the whole of the effect*. Four boundaries, each estimated with the same design:

1. It did not **break relationships** — dyad triple difference.
2. It did not **shrink the corridor** — country-pair volumes.
3. It did not **change what these investors did** — activity and all secondary outcomes.
4. It has **no identified counterpart on the British side**.

These investors did not invest less, elsewhere, differently or worse. They stopped picking the British.

It did not break relationships: the triple difference

Construction. Every dyad (i,j) of European investors co-investing at least once before 2014. *Persistent* if ≥ 2 shared deals, *occasional* otherwise. Outcome: the dyad is reactivated (shares ≥ 1 further deal) in semester s.

$1{,}347$ pairs with a British partner vs. $7{,}224$ purely continental pairs; $197{,}133$ dyad-semesters, 29 clusters.

$$\text{active}_{ds} = \beta_1(\text{pers.}_d \times \text{post}_s) + \beta_2(\text{pers.}_d \times \text{post}_s \times \text{UK}_d) + \beta_3(\text{post}_s \times \text{UK}_d) + \Gamma X_{ds} + \alpha_d + \delta_s + \varepsilon_{ds}$$

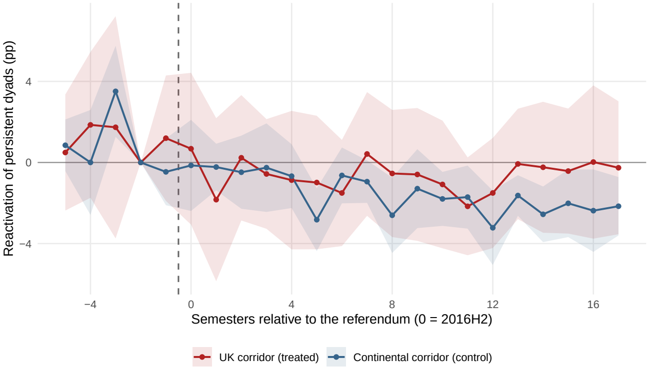
Result

$\beta_2 = +0.30$ pp, $p = 0.77$, 95% CI $[-1.84, +2.51]$ on a semi-annual reactivation base rate of 5.63%. **We exclude any reduction in the survival of British ties exceeding 1.8 pp — one third of the baseline.**

Meanwhile $\beta_1 = -2.26$ pp: persistent ties decay everywhere, at the same rate whether the partner sits in London or in Milan.

▸ How to read a triple diff

Dyad reactivation — event study



Reactivation of persistent dyads, UK corridor vs. continental corridor. Early classification (pre-2014). The curves coincide: British ties do not die faster.

This is the result that gives the paper its title: what changed is **selection**, not destruction — avoidance, not divorce.

It did not shrink the corridor

Unit. An unordered pair of European countries. **Outcome.** asinh of the deals jointly syndicated in a semester — defined *identically* for treated and control units.

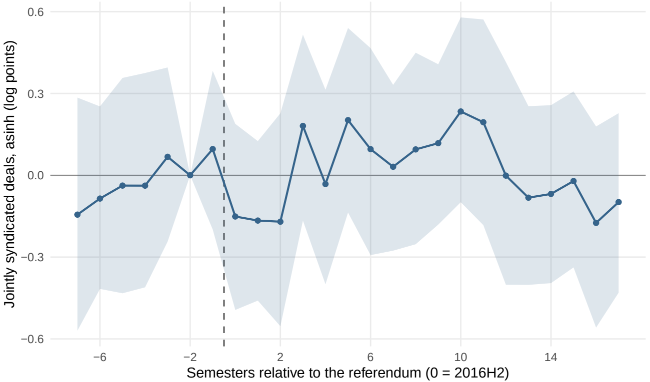
18 UK–continental corridors vs. 55 continental–continental.

Result

$+0.03$ log points, $p = 0.81$, CI $[-0.204, +0.261]$: any volume effect is bounded within $[\mathbf{-20\%}, \mathbf{+26\%}]$. Pre-trends flat ($p = 0.87$); 2014 placebo $+0.12$ ($p = 0.33$).

The two nulls together say more than either alone. Ties survive at the continental rate *and* the corridor carries the same volume. An 8.2-point shift leaving both untouched is a shift in *which* partners join new syndicates — not a contraction between the two markets.

Corridor volume — event study



Volume of jointly syndicated deals, UK–continental corridors vs. continental–continental corridors. Wild-cluster bootstrap-t, $B = 999$.

The interval also bounds the claim in the literature: increases of the order of $+80\%$ read off post-2016 data lie far outside $[-20\%, +26\%]$.

It did not change what these investors did

The conjecture. Investors cutting British partners compensate elsewhere. Small exposed VCs vs. same-size never-exposed peers: $+6.7$ pp on P(active), $p < 0.001$, clean pre-trends.

It does not survive the battery

Other corridors, identical protocol: **USA** $+7.4$, **CHE** $+5.6$, **non-EU OECD** $+7.7$ — all clearing the grid the UK does not clear. And mis-timed: $+3.2$ ($p = 0.18$) in window $[0, 6]$, the effect arriving only with the 2020–21 boom.

The source of the bias. Of 757 never-exposed small VCs, 548 (72%) never co-invested with *any* foreign European partner: the comparison contrasts networked with isolated firms.

Equally networked controls halve it to $+3.6$; isolated controls raise it to $+7.9$ — the diagnostic. **Not a precise null**: *not identified*, not zero.

The remaining outcomes

Same sample, same design, early classification, small × post control, WCB				
Outcome	Estimate	95% CI	p	Pre-trends
Deal count (asinh)	-0.03	$[-0.15, +0.08]$	0.52	ok (.53)
Early-stage share (pp)	-0.83	$[-8.30, +8.08]$	0.81	ok (.20)
Continental-target share (pp)	$+2.47$	$[-8.09, +13.38]$	0.55	ok (.24)
Follow-on round within 4y (pp)	$+0.16$	$[-8.96, +8.96]$	0.96	ok (.83)
Exit (M&A or IPO) within 4y (pp)	$+4.72$	$[-3.00, +13.35]$	0.16	ok (.13)
Not identified — fails the pre-trend test:				
Mean round size (asinh)	-0.09	$[-0.23, +0.05]$	0.21	fail (.00)

The geography result is the one a reader expects to move: exposed investors did *not* redirect deal flow towards continental targets ($+2.5$ pp, $p = 0.55$), nor towards American ones. Mean round size is reported without interpretation — the groups were already diverging.

$\Rightarrow$ We list every outcome examined, so that the selection of reported results can be verified rather than trusted.

No identified counterpart on the British side

A British panel returns a mirror estimate of -6.0 pp ($p = 0.039$) on the continental aggregate.

The protocol holds the universe fixed and varies the corridor: it asks of a British VC what it asks of a continental one. Here the prediction is *sharper* — EU corridors should move, non-EU corridors should not.

The prediction is not merely unmet — it is inverted

- Continental aggregate: reproduces, but fails both pre-trend tests ($p = 0.04$, $p = 0.03$).
- **USA: clears the grid outright** — -5.0 pp, pre-trends 0.95 / 0.97, -7.1 in 2022–25.
- CHE: -10.9. FRA, DEU, NLD: null.

Stated narrowly. 7 of 14 corridors dropped, nulls resting on 40–50 firms: the British side is *not identified*, which is not “nothing happened”.

What can be eliminated

A shift in partner choice, dated to a vote, sustained nine years, with no change in volumes, relationships, activity or outcomes, where no rule was ever binding. What produced it?

One class of explanation is eliminated.

- **Not compliance.** No rule governing syndication changed, and the response precedes the only legal change by four and a half years.
- **Not the opportunity set.** Restricting to companies whose first round predates June 2016 leaves the pattern intact ($+5.8$ pp, $p = 0.002$; 2014 placebo $p = 0.87$).
- **Not a general retreat from foreign partners.** USA, CAN, JPN, FRA, DEU, NLD and non-EU OECD exposure all unchanged.

What cannot be adjudicated

We set the remaining channels out rather than choose between them.

- **Anticipation.** Frictions priced in advance. A syndicate formed in 2017 is a commitment extending well past 2020.
- **Coordination cost.** Uncertainty raises the cost of *entering* a tie, not of holding one — exactly the pattern observed.
- **Boundary realignment.** A vote makes a boundary salient and networks realign on it, with no agent facing a changed constraint.
- **An upstream constraint.** European public institutions are substantial LPs; a tightening of their geographic mandates would bind above the investor.

The last is the most serious alternative and **we cannot test it**: Crunchbase records investors, not their LPs. Well defined with fund-level LP data — flagged as the next step, not ruled out.

Integration rests on behaviour, not only on rules

The legal architecture was, on this margin, entirely undisturbed. If integration here were sustained by the framework, nothing should have happened.

Something did: a forty-four percent relative decline, beginning at the vote, still present nine years later.

The implication runs both ways — the second is the uncomfortable one

Dismantling the framework was not necessary to disintegrate the network: an expectation sufficed. Symmetrically, restoring it will not be sufficient to reintegrate it.

For the Capital Markets Union agenda: instruments that operate on rules address the part of integration that, here, was never the binding one.

A note on identification under market-wide shocks

The DiD literature (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021) presumes *some* units are untreated. Here none are.

Three coefficients that were large, signed, significant — and spurious:

1. a dyadic divorce of -7.1 pp: regression to the mean, reproduces for FRA and DEU;
2. an activity expansion: reproduces for USA and CHE exposure;
3. a symmetric British rupture: dissolves under early classification.

None was removed by a better estimator, but by a contemporaneous counterfactual.

One observation for practitioners

The date placebo was the stronger safeguard, the pre-trend test the weaker one. Several results passed at $p > 0.9$ and died on a 2014 placebo. The default order is the reverse.

Takeaways

Nothing restricted co-investment; the syndication changed anyway.

- **The fact.** -8.2 pp ($p < 0.001$; CI $[-13.1, -3.2]$) on a base of 18.7%, dated on the referendum, still present in 2022–25.
- **The boundaries.** Ties survive ($+0.3$ pp, $p = 0.77$); the corridor holds its volume ($[-20\%, +26\%]$); counts, stages, geography, amounts, follow-ons and exits are null.
- **The specificity.** Of thirteen corridors, two clear the grid — and only the British one declines *from* the referendum.
- **The channel.** Not identified, and not asserted.

A framework left intact did not preserve the behaviour it is credited with sustaining — so the rules were not the binding constraint.

Thank you

Appendix: why the AIFMD passport is not the answer

- Syndication is **not a passported activity**: taking an equity stake is not conditioned on Union membership.
- Two funds may invest in the same round without notifying anyone, without a common licence.
- The passport governs the **management and marketing of funds to professional investors** — raising capital, not deploying it. It held for the UK until end-2020 anyway.

Three consequences.

1. No mechanical explanation: no rule required the shift, at any date.
2. Timing is informative: a legal mechanism should bite in 2020, four and a half years after the response.
3. The shock is composite, and we do not decompose it.

Back

Appendix: how to read a triple difference

A DiD compares *before/after* across *treated/control*. A triple difference adds a third dimension, so that the second difference itself gets a control.

Δ post – pre in P(reactivation)	Persistent ties	Occasional ties
UK–continental dyads	a	b
continental–cont. dyads	c	d

$$\beta_2 = \underbrace{(a-b)}_{\text{decay of persistent UK ties}} - \underbrace{(c-d)}_{\text{decay of persistent continental ties}}$$

- $(a-b)$ nets out anything hitting *all* UK dyads after 2016 (a common UK shock).
- $(c-d)$ nets out the *natural* decay of persistent ties, which happens everywhere ($\beta_1 = -2.26$ pp).
- β_2 is what remains: the excess mortality of *persistent British* ties. It is $+0.30$ pp, $p = 0.77$.

Identifying assumption: the *difference* between persistent and occasional ties would have evolved in parallel across the two corridors absent Brexit — weaker than requiring each cell to be parallel.

Back

Appendix: precision — what the nulls exclude

		Estimate	95% CI	Base rate	MDE
Effect	UK-partner rate (pp)	-8.20	$[-13.11, -3.15]$	18.7%	6.64
Null	Dyad survival, triple diff. (pp)	$+0.30$	$[-1.84, +2.51]$	5.6%	2.62
Null	Corridor volume (log pts)	$+0.03$	$[-0.20, +0.26]$	7.5 deals	0.32
Null	Activity, window $[0, 6]$ (pp)	$+3.18$	$[-1.47, +8.16]$	26.0%	6.17
Not id.	British side, mirror estimate (pp)	-6.01	$[-11.52, -0.33]$	22.5%	8.51

- **Corridor:** decisive. Bounds the volume effect within $[-20\%, +26\%]$; increases of the order of $+80\%$ read off post-2016 data lie far outside it.
- **Dyad:** informative. Excludes any reduction in semi-annual survival beyond 1.8 pp, one third of the 5.6% baseline.
- **Activity:** *not* precise — admits $+8.2$ pp. Our claim rests on the falsification battery, not on precision.
- **British side:** not a null — its interval excludes zero. Disqualified by pre-trends and placebos, and its MDE (8.5) exceeds the effect it reports (6.0).

Back

Appendix: what Crunchbase does not see

Coverage follows visibility, not a sampling frame.

- Thinner at the earliest stages and outside the large western markets; only investors recorded as organisations with a HQ country.
- Our sample: continental VCs with ≥ 2 pre-2014 deals and a pre-2014 British partner — firms visible enough to appear at least three times.

$\Rightarrow$ The result describes **established, observable** investors. If the effect were concentrated in the small, informal or peripheral tail, this design would not find it.

What it does not contain. The investors in a round, not the LPs behind them — so the upstream channel is untestable here. A limit of the source, not of the design.

Back

Appendix: Ireland

Ireland is the only country removed by name, so it owes the reader a defence. We test it rather than argue it.

Definition of CE	Universe	Estimate	Pre-trends
Excluding Ireland **[main]**	448	-8.20	ok (.23 / .36)
Including Ireland	470	-8.01, CI $[-12.62, -3.26]$	ok (.11 / .18)
Ireland alone	22	-6.8, CI $[-49.8, +29.7]$	uninformative

- **Substantive reason.** The land border and the Common Travel Area survived, and Dublin took entities leaving London.
- **Verdict.** Immaterial: a reader preferring the wider definition may read -8.0 for -8.2 and nothing else follows.
- **Caveat.** Run on the main estimate only: the dyad, corridor and UK panels were not re-estimated under the wide definition.

Appendix: the pre-existing opportunity set

The alternative reading. If fewer British-linked ventures were founded after 2016, exposed investors would take fewer British partners without *choosing* to.

We fix the opportunity set. Restricting the panel to portfolio companies whose first funding round predates June 2016 — firms that existed, and were fundable, before the vote:

Fixed pre-2016 pool	Estimate	Pre-trends
Activity	$+5.8$ pp ($p = 0.002$)	ok (.80)
2014 date placebo, same pool	$+0.3$ pp ($p = 0.87$)	—

$\Rightarrow$ The pattern is not produced by the entry of new firms into the sample.

Reported as a boundary on the interpretation, *not* as support for the activity result — which we decline to identify for the independent reasons of the battery.